Winning Space Race with Data Science

SpaceX Falcon 9 First-Stage Landing Prediction

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IBM Applied Data Science Capstone · October 2026

https://github.com/jegathesh311-tech/AppliedDataScienceCapstone

Outline

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- Introduction
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  - Insights from EDA (visualization and SQL)
  - Launch site proximity analysis (Folium)
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Executive Summary

- **Methods:** collected launch data from the SpaceX REST API and scraped Falcon 9 records from Wikipedia; cleaned it and built a binary landing label
  - Explored with Pandas/Seaborn charts and 10 SQL queries on SQLite; mapped sites with Folium; built a Plotly Dash dashboard
  - Tuned Logistic Regression, SVM, Decision Tree and KNN with GridSearchCV (10-fold CV) and compared them on a held-out test set
- **Results:** 90 Falcon 9 launches analysed; overall landing success rate 66.7% (60 of 90)
  - Success climbed from 0% (2010–2013) to 90% in 2019; flight experience is the strongest signal
  - KSC LC-39A has the best site success rate (76.9%, 10 of 13 in the dashboard data)
  - ES-L1, GEO, HEO and SSO orbits: 100% success; GTO is the hardest common orbit (51.9%)
  - All four models score 83.3% test accuracy; the Decision Tree has the best validation accuracy (87.5%)

Introduction

- **Background:** SpaceX lists Falcon 9 launches at about $62 million, while other providers charge $165 million or more
  - Most of the saving comes from landing and reusing the first stage
- **Problem:** a new launch company, Space Y, wants to bid against SpaceX
  - If we can predict whether the first stage will land, we can estimate the real cost of a SpaceX launch
- **Questions this project answers:**
  - How do launch site, payload mass, orbit, flight number and year affect landing success?
  - Where are the launch sites, and what are they close to?
  - Can a machine-learning model predict a successful landing, and how accurately?

Section 1

Methodology

Methodology

- **Data collection:** SpaceX REST API (api.spacexdata.com/v4) and web scraping of the Wikipedia "List of Falcon 9 and Falcon Heavy launches" page
- **Data wrangling:** kept Falcon 9 launches only, filled missing payload mass with the mean, turned landing outcomes into a Class label (1 = landed, 0 = not)
- **Exploratory data analysis:** scatter, bar and line charts with Pandas/Seaborn; SQL queries on SQLite
- **Interactive visual analytics:** Folium map of sites, outcomes and proximities; Plotly Dash dashboard
- **Predictive analysis:** standardised features, 80/20 train/test split, four classifiers tuned with GridSearchCV (cv = 10), compared on test accuracy, Jaccard and F1

Data Collection – SpaceX API



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/1_Data_Collection.ipynb

Data Collection – Web Scraping



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/2_Web_Scraping.ipynb

Data Wrangling

- Checked missing values: LandingPad is 28.9% null (no pad used); all other columns complete
- Counted launches per site with value_counts(): CCAFS SLC 40 = 55, KSC LC 39A = 22, VAFB SLC 4E = 13
- Counted orbits: GTO 27, ISS 21, VLEO 14, PO 9, LEO 7, SSO 5, MEO 3, others 1 each
- Grouped outcomes: True ASDS 41, None None 19, True RTLS 14, False ASDS 6, True Ocean 5, False Ocean 2, None ASDS 2, False RTLS 1
- Defined bad outcomes {False ASDS, False Ocean, False RTLS, None ASDS, None None} → Class = 0, otherwise Class = 1
  - Result: 60 landed, 30 did not → success rate 66.7%
- One-hot encoded Orbit, LaunchSite, LandingPad and Serial for modelling (83 feature columns)

Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/3_Data_Wrangling.ipynb

EDA with Data Visualization

- **Scatter plots (hue = Class)** – show how two variables relate to the landing outcome:
  - Flight Number vs Payload Mass, Flight Number vs Launch Site, Payload Mass vs Launch Site
  - Flight Number vs Orbit, Payload Mass vs Orbit
- **Bar chart** – mean success rate for each orbit type, to compare categories
- **Line chart** – yearly average success rate, to show the trend over time
- **Feature engineering** – selected 12 features and one-hot encoded the categorical ones for the models

Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

EDA with SQL

- Loaded Spacex.csv into SQLite (table SPACEXTABLE, blank rows removed) and queried it with %sql magic
- Queries performed:
  - Unique launch site names; 5 records where the site begins with "CCA"
  - Total payload carried for NASA (CRS); average payload for booster F9 v1.1
  - Date of the first successful ground-pad landing
  - Boosters with a drone-ship success and payload between 4,000 and 6,000 kg
  - Count of successful and failed mission outcomes
  - Boosters that carried the maximum payload (subquery)
  - 2015 drone-ship landing failures by month (substr on Date)
  - Ranking of landing outcomes between 2010-06-04 and 2017-03-20

Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/4_Exploratory_Data_Analysis_Using_SQL.ipynb

Build an Interactive Map with Folium

- **Circles and text markers** at each launch site (and NASA JSC as a reference) to show where the sites are
- **MarkerCluster** of every launch, with green markers for successful landings and red for failures, so each site's record is visible at a glance
- **MousePosition** plugin to read coordinates of nearby features
- **Distance markers and PolyLines** from CCAFS SLC-40 to the nearest coastline, highway, railway and city, using the haversine formula
- Purpose: understand why sites are where they are, and whether location relates to success

Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/6_Interactive_Visual_Analytics_with_Folium_Lab.ipynb

Build a Dashboard with Plotly Dash

- **Launch-site dropdown** (All Sites or a single site) drives both charts
- **Pie chart:** for All Sites, each site's share of total successful launches; for one site, its success (1) vs failure (0) split
- **Payload range slider** (0–10,000 kg) to filter launches by payload
- **Scatter chart:** payload mass vs landing outcome, coloured by booster version category
- Purpose: let a user find the best site, payload range and booster version interactively

Dash app: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/7_Interactive_Visual_Analytics_and_Dashboard_with_Plotly_Dash.py

Predictive Analysis (Classification)



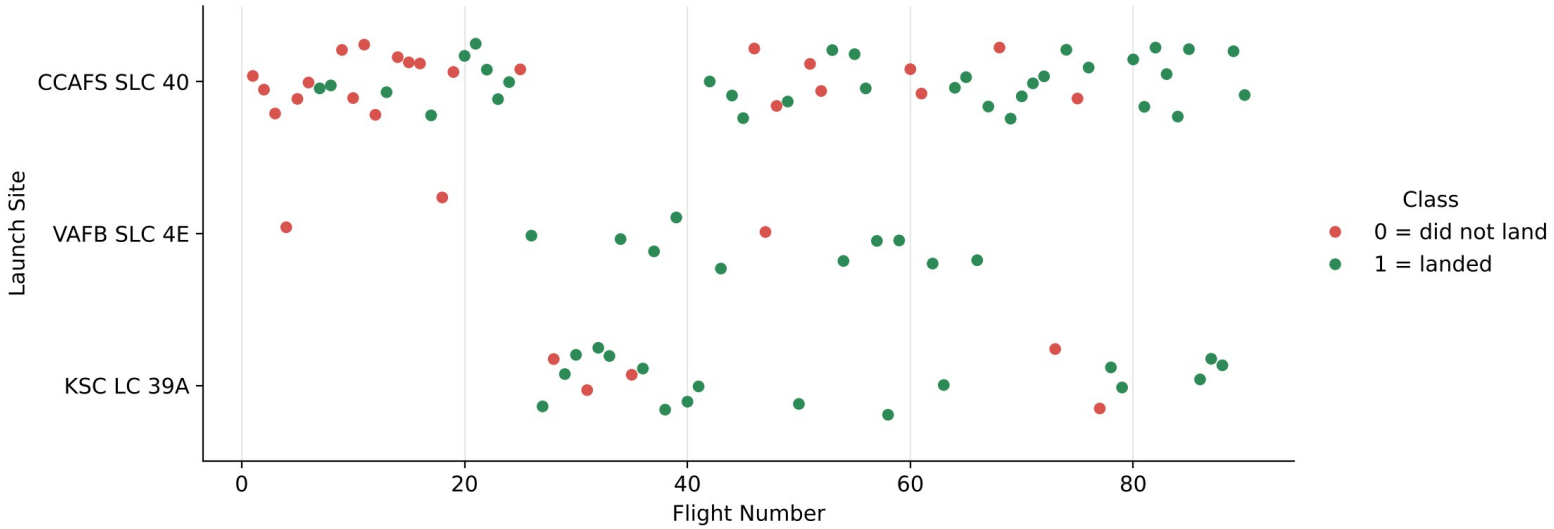
Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/8_Machine_Learning_Predictive_Analysis.ipynb

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

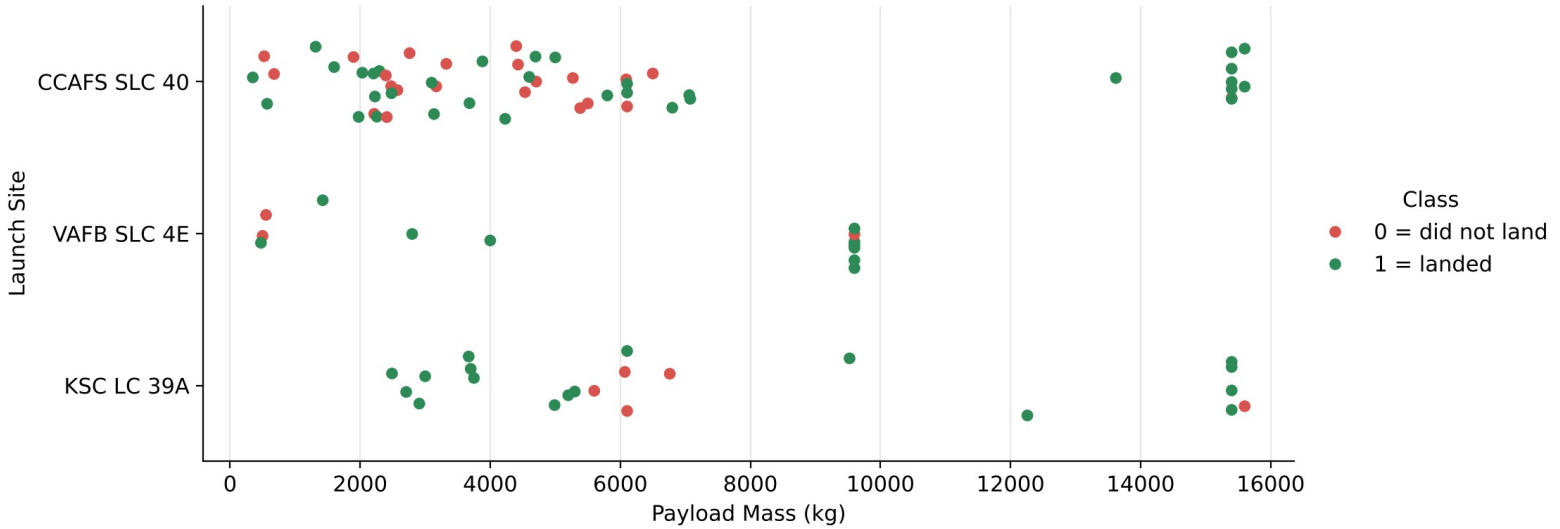
- CCAFS SLC 40 hosts most launches (55 of 90), including nearly all early flights
- Early flights at every site were mostly failures
- Success becomes the norm as flight number rises – SpaceX learned with each launch
- VAFB SLC 4E and KSC LC 39A were used later, so they show higher success



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

Payload vs. Launch Site

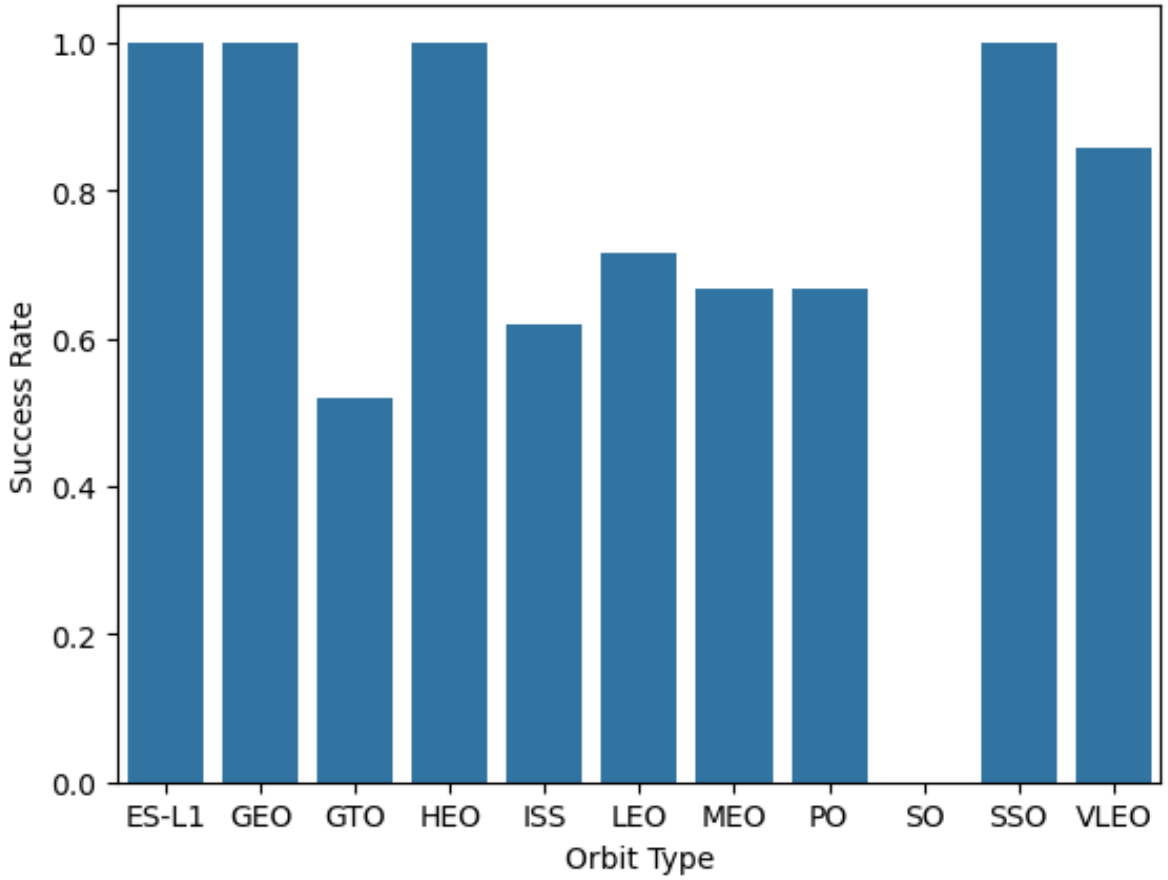
- Most launches carry under 8,000 kg
- Heavy payloads (above ~9,000 kg) mostly landed successfully
- VAFB SLC 4E has no launches above ~10,000 kg
- KSC LC 39A has all its successes below ~5,500 kg plus heavy VLEO flights



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

Success Rate vs. Orbit Type

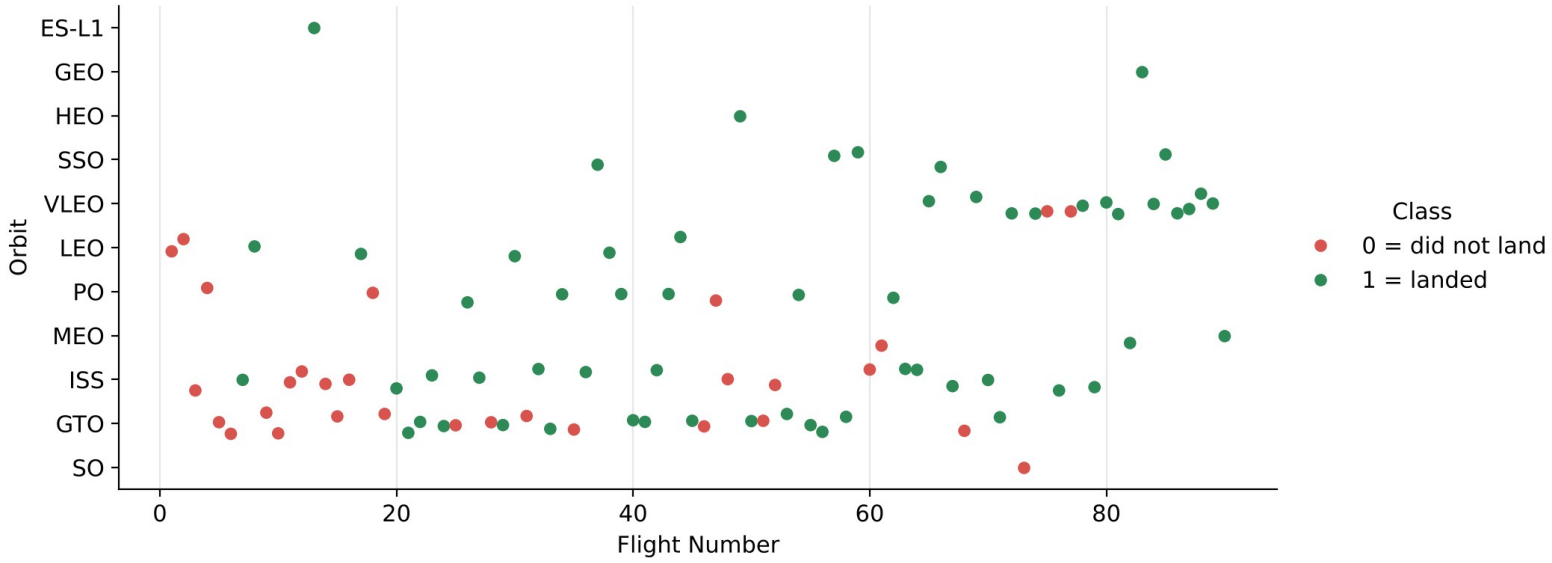
- ES-L1, GEO, HEO and SSO: 100% success (but only 1–5 launches each)
- VLEO 85.7% (14 launches) is the best large orbit group
- GTO is the lowest of the common orbits at 51.9%
- SO: 0% (a single launch)



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

Flight Number vs. Orbit Type

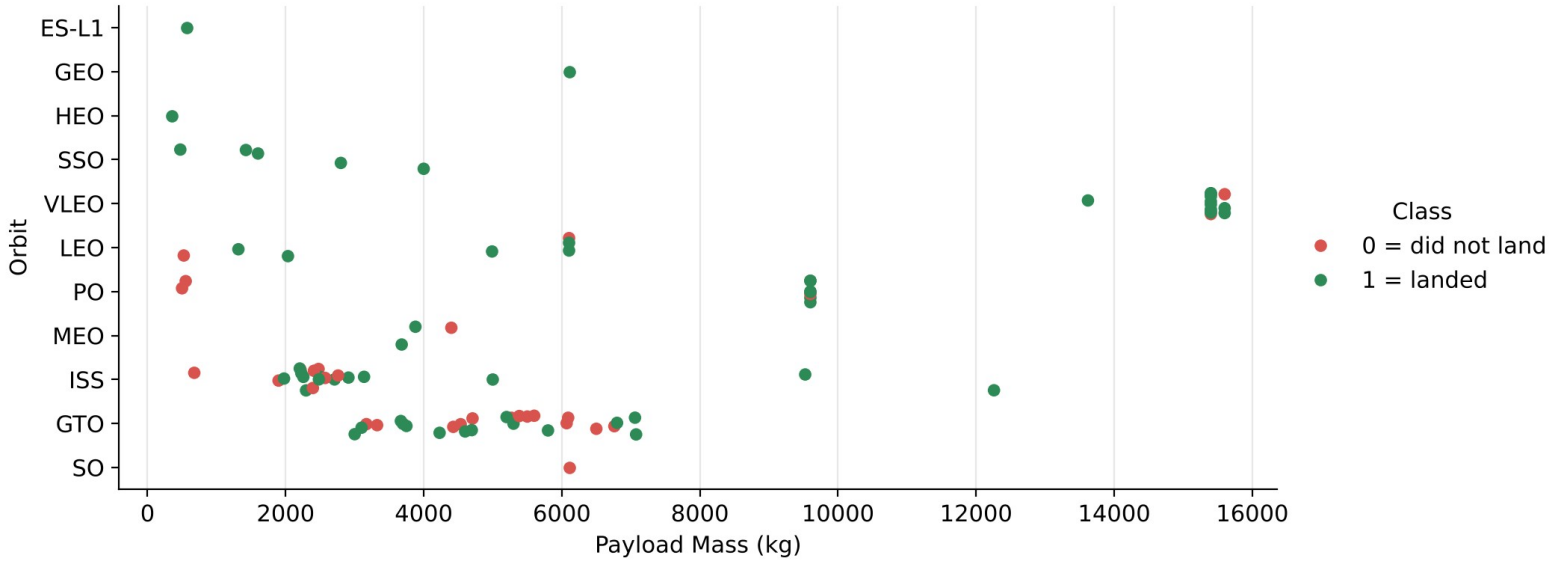
- LEO: success rises with flight number
- GTO: no clear link between flight number and success
- VLEO launches all come late (flight 65+) and mostly succeed
- Orbits sorted by success rate (top = highest)



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

Payload vs. Orbit Type

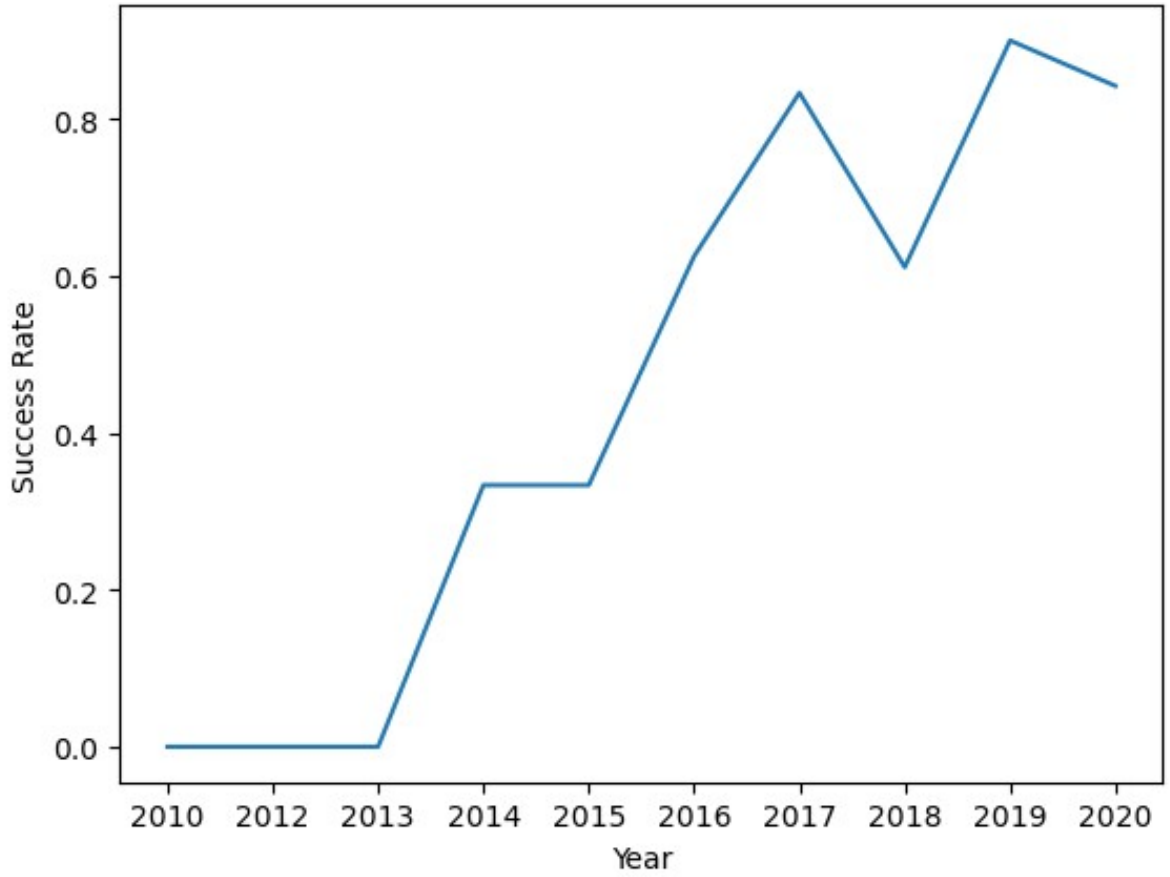
- Polar (PO), LEO and ISS: heavier payloads land successfully more often
- GTO: successes and failures are mixed across all payloads
- VLEO carries the heaviest payloads (~15,600 kg) and mostly succeeds



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

Launch Success Yearly Trend

- Success rate was 0% from 2010 to 2013
- It rose steadily from 2013: 33% (2014–15), 62% (2016), 83% (2017)
- A dip to 61% in 2018, then 90% in 2019 and 84% in 2020



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

Section 3

EDA with SQL – Results

All Launch Site Names

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Result

Four unique launch sites: three in Florida and one in California (VAFB).

Launch Sites Beginning with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

Date	Booster	Site	Payload	kg	Orbit	Customer	Mission	Landing
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Result

First five CCA launches: all F9 v1.0 from CCAFS LC-40; no landing succeeded.

Total Payload Mass – NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
     WHERE Customer = 'NASA (CRS)'
```

SUM(PAYLOAD_MASS__KG_)
45,596

Result

Boosters carried a total of 45,596 kg for NASA (CRS).

Average Payload Mass – F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
     WHERE Booster_Version = 'F9 v1.1'
```

AVG(PAYLOAD_MASS__KG_)
2,928.4

Result

Booster version F9 v1.1 carried 2,928.4 kg on average.

First Successful Ground Pad Landing

```
%sql SELECT Date FROM SPACEXTABLE
     WHERE Landing_Outcome = 'Success (ground pad)' ORDER BY Date LIMIT 1
```

Date
2015-12-22

Result

The first successful ground-pad landing was on 22 December 2015.

Successful Drone Ship Landing with Payload 4,000–6,000 kg

```
%sql SELECT Booster_Version FROM SPACEXTABLE
     WHERE Landing_Outcome = 'Success (drone ship)'
     AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Result

Four boosters, all Full Thrust (FT) versions; two were reused boosters (.2).

Total Successful and Failed Mission Outcomes

```
%sql SELECT Mission_Outcome, COUNT(Mission_Outcome)
     FROM SPACEXTABLE GROUP BY Mission_Outcome
```

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success (trailing space in data)	1
Success (payload status unclear)	1

Result

100 of 101 missions succeeded; only one failed in flight. (Mission success ≠ booster landing.)

Boosters that Carried the Maximum Payload

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ =
     (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
```

Booster_Version	Booster_Version	Booster_Version
F9 B5 B1048.4	F9 B5 B1048.5	F9 B5 B1058.3
F9 B5 B1049.4	F9 B5 B1051.4	F9 B5 B1051.6
F9 B5 B1051.3	F9 B5 B1049.5	F9 B5 B1060.3
F9 B5 B1056.4	F9 B5 B1060.2	F9 B5 B1049.7

Result

Twelve launches tie for the maximum payload; all used reused Block 5 (B5) boosters.

2015 Launch Records: Drone Ship Failures by Month

```
%%sql SELECT CASE substr(Date,6,2) ... END AS Month, Landing_Outcome,
       Booster_Version, Launch_Site FROM SPACEXTABLE
 WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date,0,5) = '2015'
```

Month	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Result

Two drone-ship failures in 2015 (January and April), both F9 v1.1 from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql SELECT Landing_Outcome, COUNT(Landing_Outcome) FROM SPACEXTABLE
 WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
 GROUP BY Landing_Outcome ORDER BY COUNT(Landing_Outcome) DESC
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Result

"No attempt" was most common in this early period; drone-ship landings were a 5-to-5 split.

Section 4

Launch Sites Proximities Analysis

All Launch Sites on a Map

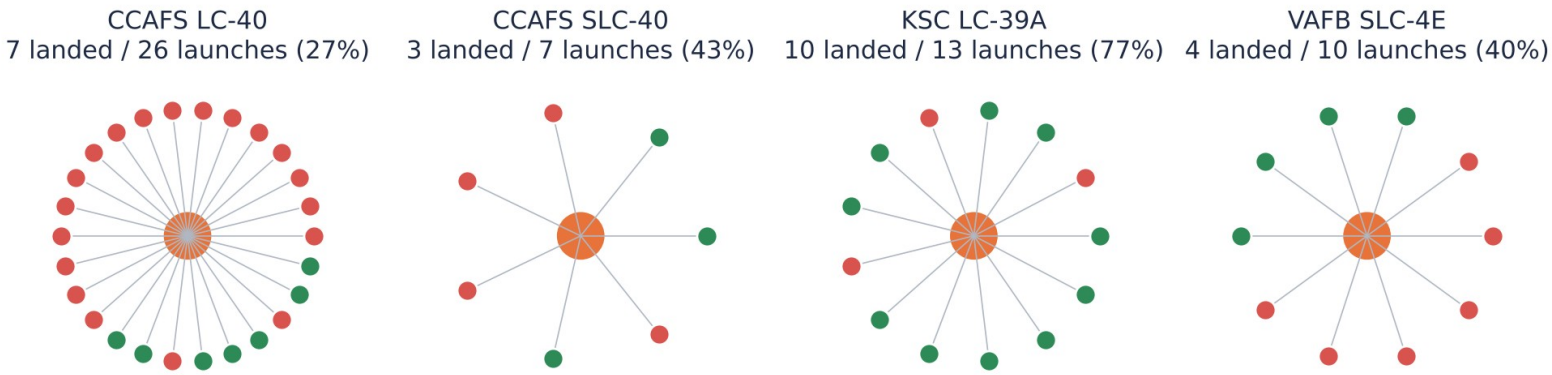
- All four sites are in the USA: three on Florida's east coast, one on the California coast
- Every site sits right on the coast, so launches and failed boosters head out over the ocean
- Sites are as far south as practical – Earth's rotation is fastest near the equator, which helps reach orbit



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/6_Interactive_Visual_Analytics_with_Folium_Lab.ipynb

Launch Outcomes per Site (Marker Clusters)

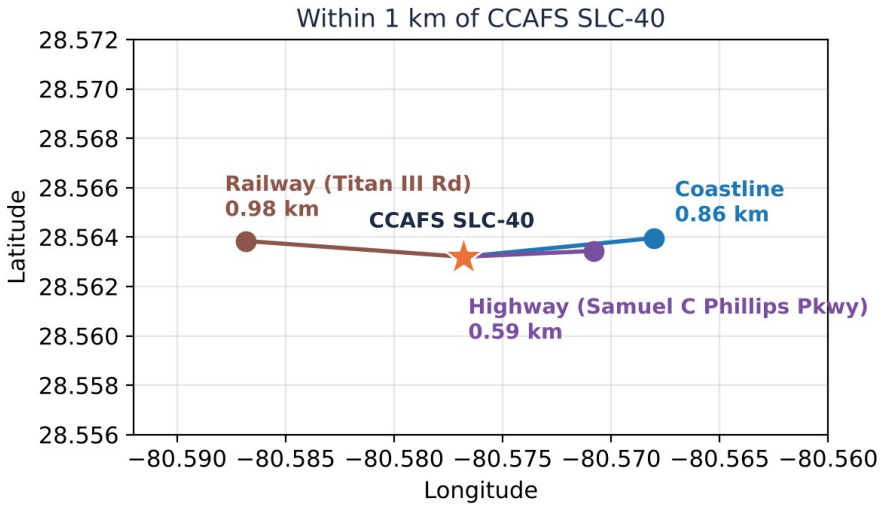
- Green = successful landing, red = failed (class column)
- KSC LC-39A has the highest share of green: 10 of 13 (77%)
- CCAFS LC-40 has the most launches (26) but only 27% landed – mostly early flights

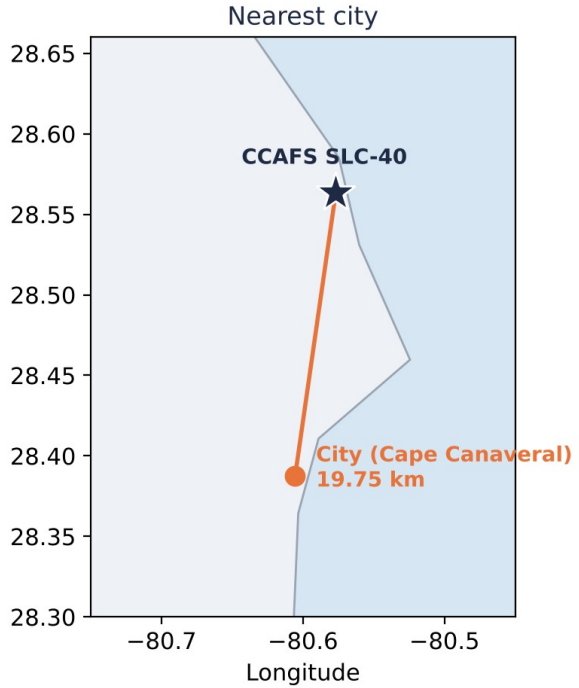


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Proximity of CCAFS SLC-40 to Nearby Features

- Highway (Samuel C Phillips Pkwy): 0.59 km
- Coastline: 0.86 km
- Railway (Titan III Rd): 0.98 km
- Nearest city (Cape Canaveral): 19.75 km
- Sites are close to transport links and the sea, but kept well away from towns for safety





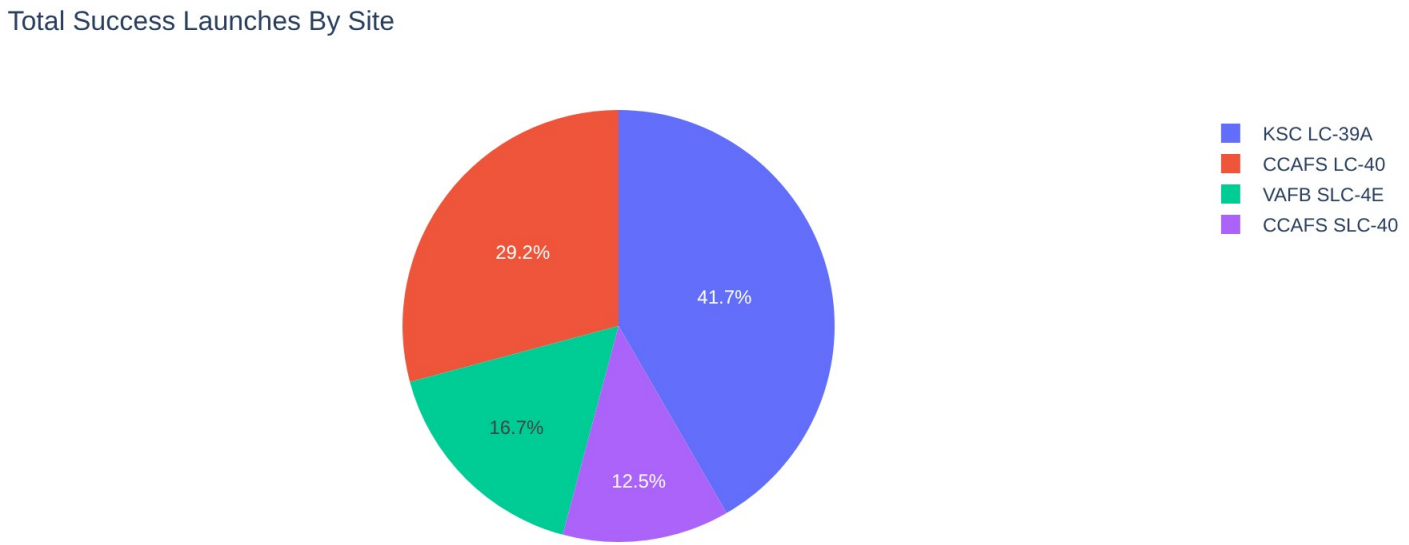
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Section 5

Build a Dashboard with Plotly Dash

Launch Success Count for All Sites

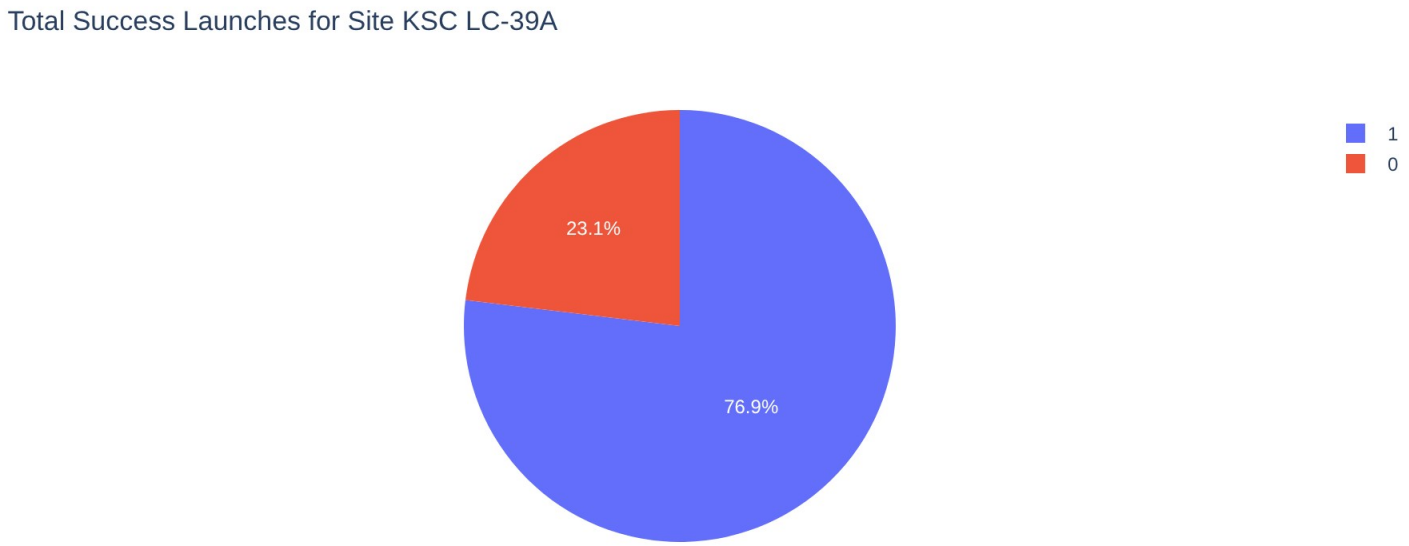
- KSC LC-39A contributes the largest share of successful launches: 41.7%
- CCAFS LC-40 follows with 29.2%, VAFB SLC-4E 16.7%
- CCAFS SLC-40 contributes the smallest share: 12.5%



Dash app: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/7_Interactive_Visual_Analytics_and_Dashboard_with_Plotly_Dash.py

Site with the Highest Launch Success Ratio

- KSC LC-39A: 76.9% success (10 of 13 launches)
- Only 3 of its 13 launches failed (23.1%)
- This is the highest success ratio of the four sites



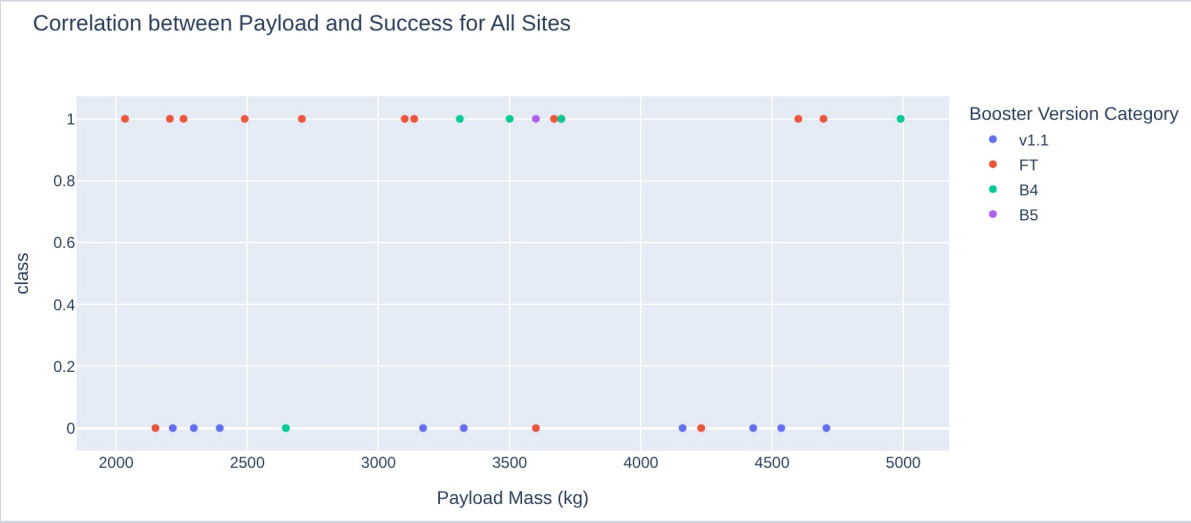
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Payload vs. Launch Outcome – All Sites

- Full range (0–10,000 kg): 2,000–4,000 kg has the best success rate (62%); 6,000–8,000 kg had 0 of 4
- Booster FT has the most successes (16 of 24); v1.1 the most failures (14 of 15 failed)



Slider at full payload range



Slider narrowed to 2,000–5,000 kg

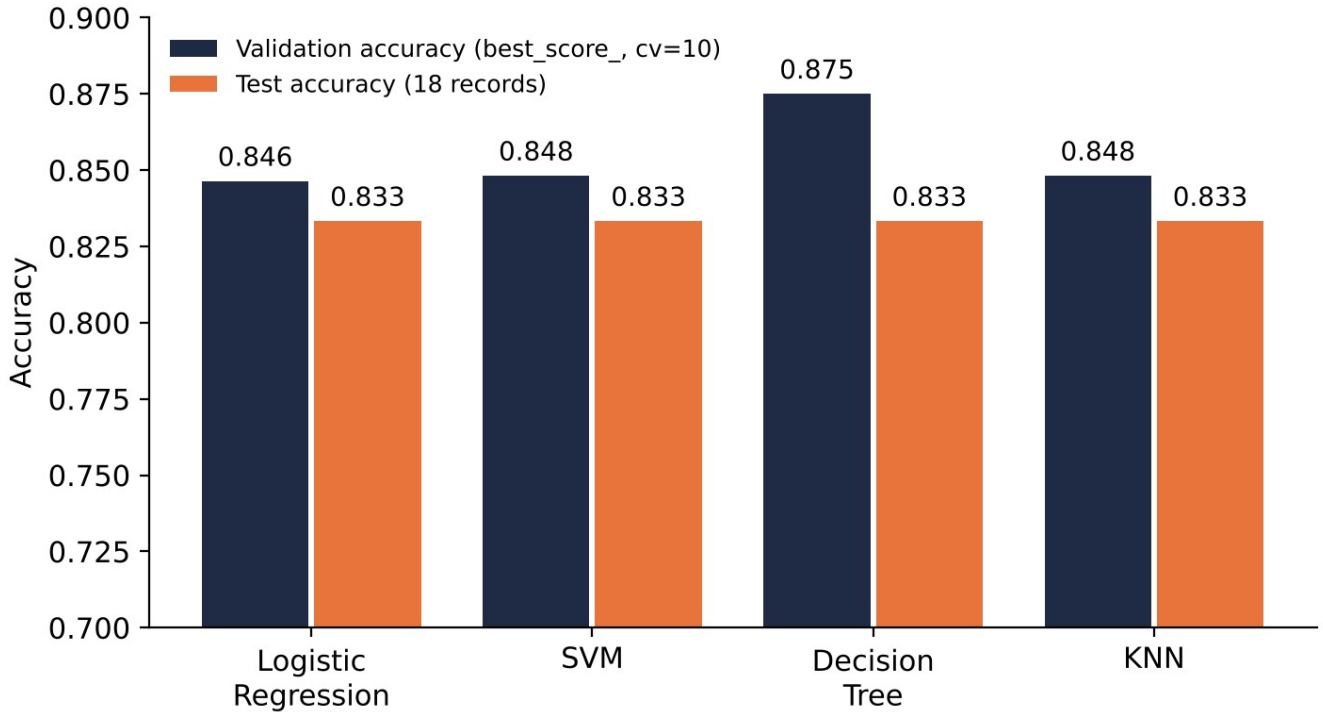
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Section 6

Predictive Analysis (Classification)

Classification Accuracy

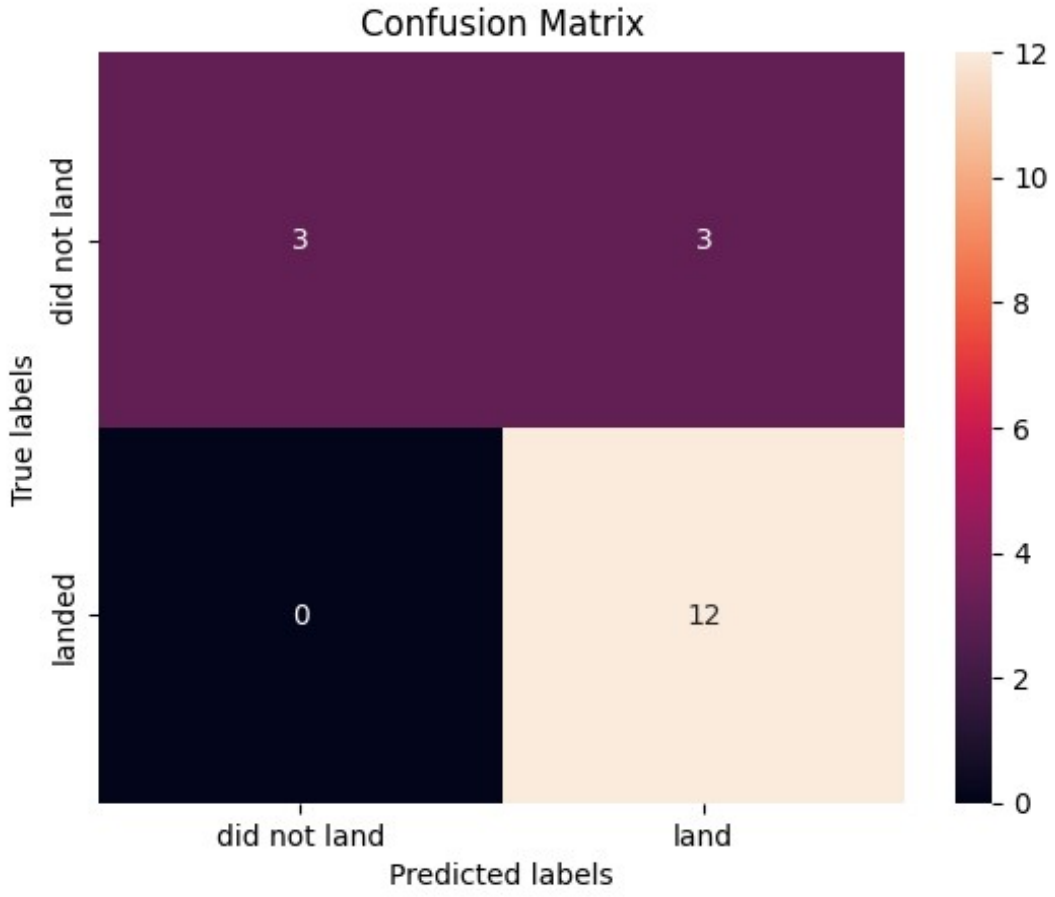
- Test accuracy on 18 records is identical for all four models: 83.3% (Jaccard 0.80, F1 0.889)
- Validation accuracy (10-fold CV) separates them:
  - Decision Tree 0.875 · SVM 0.848 · KNN 0.848 · Logistic Regression 0.846
- Best SVM kernel: sigmoid (C = 1.0, gamma ≈ 0.032)
- **Decision Tree is the best model** on validation accuracy



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/8_Machine_Learning_Predictive_Analysis.ipynb

Confusion Matrix

- Decision Tree on the 18 test records (the other three models give the same matrix)
- All 12 actual landings predicted correctly (recall = 1.00)
- Main problem: false positives – 3 of 6 failed landings predicted as "landed" (precision = 0.80)
- With only 18 test samples, one error moves accuracy by 5.6 points



Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/8_Machine_Learning_Predictive_Analysis.ipynb

Conclusions

- Landing success rose from 0% (2010–2013) to around 90% by 2019: experience matters most
- KSC LC-39A has the best success ratio (76.9%); CCAFS SLC 40 has the most launches
- ES-L1, GEO, HEO and SSO orbits reached 100% success; GTO is the hardest common orbit
- Light-to-medium payloads (2,000–4,000 kg) and FT / B4 / B5 boosters land most often
- Launch sites are near the coast and transport links, and far from cities
- A tuned Decision Tree predicts landing outcome with 83% test accuracy (87.5% validation), so Space Y can estimate SpaceX's true launch cost before bidding
- Next step: retrain on newer launches and a larger test set to separate the models more reliably

Appendix

GitHub repository: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone

1 · Data collection (SpaceX API): https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/1_Data_Collection.ipynb

2 · Web scraping: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/2_Web_Scraping.ipynb

3 · Data wrangling: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/3_Data_Wrangling.ipynb

4 · EDA with SQL: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/4_Exploratory_Data_Analysis_Using_SQL.ipynb

5 · EDA with visualization:
https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

6 · Folium map: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/6_Interactive_Visual_Analytics_with_Folium_Lab.ipynb

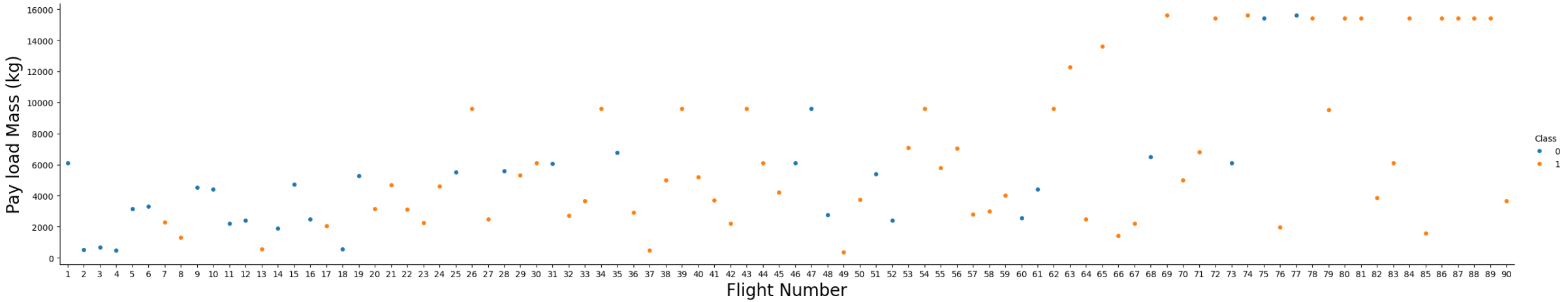
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8 · Machine learning prediction:
https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/8_Machine_Learning_Predictive_Analysis.ipynb

Notes: Folium maps are redrawn from the notebook's coordinates without base-map tiles. Dashboard images are screenshots of the Dash app run on spacex_launch_dash.csv. In the Decision Tree grid, max_features='auto' is not supported by current scikit-learn, so only 'sqrt' was searched.

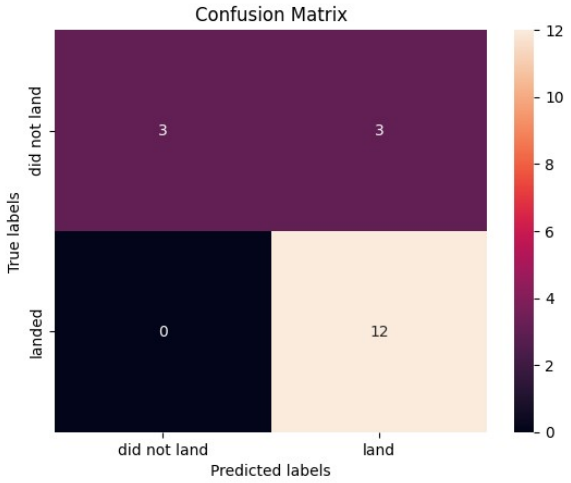
Appendix – Flight Number vs. Payload Mass

- Later flights succeed even with heavier payloads
- Early flights failed at almost every payload

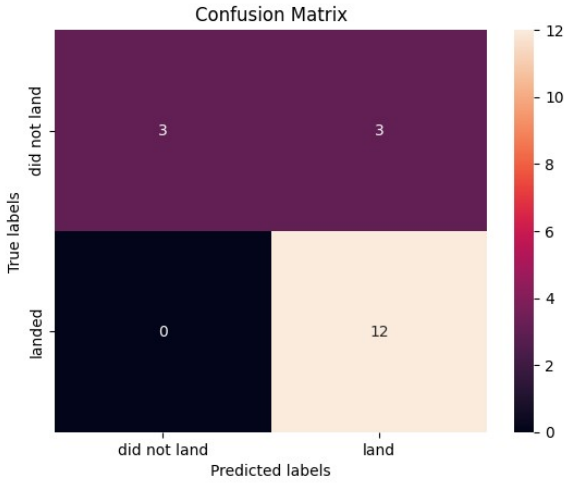


Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/5_Exploratory_Data_Analysis_Using_Pandas_and_Matplotlib.ipynb

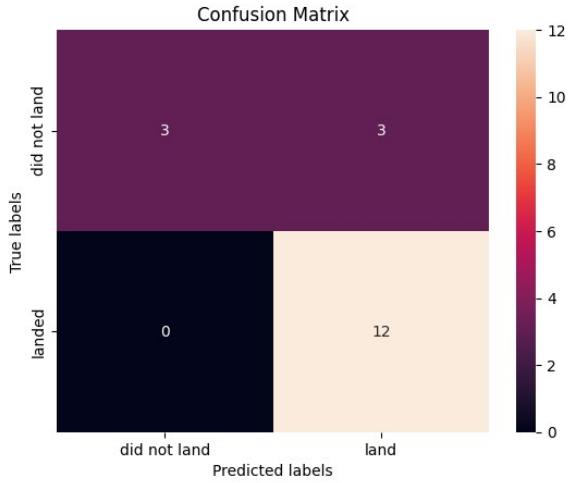
Appendix – Confusion Matrices of All Models



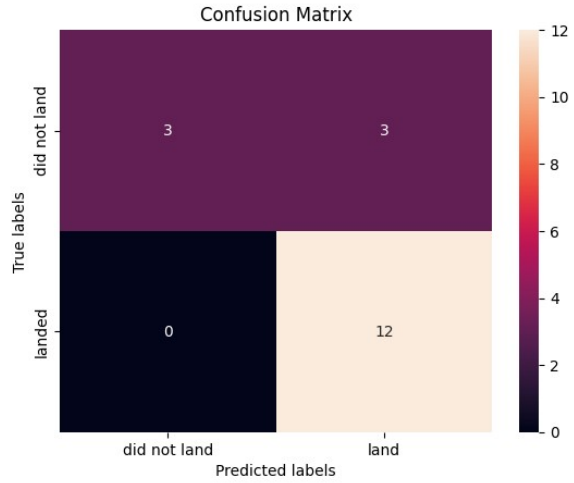
Logistic Regression



SVM



Decision Tree



KNN

Notebook: https://github.com/jegathesh311-tech/AppliedDataScienceCapstone/blob/main/8_Machine_Learning_Predictive_Analysis.ipynb